

KarthEEK Mukkavilli

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Education

University of Texas at Austin

Austin, TX

B.S. Electrical & Computer Engineering — GPA: 3.95/4.0 — Expected May 2028

Coursework: Algorithms · Computer Architecture · Operating Systems · Discrete Mathematics · Software Design · Digital Logic Design

Elements of Computing Certificate (CS) · Business Foundations Minor

Technical Skills

Languages: Python, C, C++, Java, TypeScript, SQL

Frameworks/Tools: React, FastAPI, Spring Boot, Node.js, Docker, GitLab CI/CD, SQLAlchemy, Git

Concepts: REST APIs, Microservices, OAuth 2.0, CI/CD, RAG Pipelines, Agentic Systems

Experience

Southwest Research Institute

San Antonio, TX

Student Engineer, Tactical Aerospace Department (Division 16)

May 2026 – Present

- Modernizing a legacy firmware build pipeline in an ITAR-controlled environment, consolidating fragmented codebases into a single validated, version-controlled monorepo.
- Validated build reproducibility across multiple development machines ahead of full containerization, eliminating environment-specific toolchain drift.
- Containerized the build toolchain with Docker and integrated it into a GitLab CI/CD pipeline, replacing a manual, GUI-driven workflow with automated, reproducible builds.

Human Centered Robotics Laboratory — UT Austin

Austin, TX

Research Assistant

Jan 2026 – Present

- Integrated a PyTorch Transformer to predict 16-DOF bionic hand motion from glove telemetry, achieving 1ms sensor-to-actuation latency.
- Refined a modular PID motor control API for 10 joints, improving tracking from unreliable to consistently stable real-time replication.
- Debugged and profiled the sensor-to-actuation pipeline end-to-end to isolate and resolve latency bottlenecks.

National Oilwell Varco

Houston, TX

Embedded Systems Intern

Jun – Aug 2025

- Replaced a \$20K/site legacy PLC system with a custom nRF9160 board under \$250 (98% cost reduction), deployed across 50 field units.
- Engineered Zephyr RTOS C firmware for LTE-M/GNSS over a 40-node LoRa mesh to AWS IoT Core, writing a safe-shutdown protocol to prevent LTE tower rejection on power cycle.
- Automated TLS certificate provisioning via Python + AT+CMNG scripting, reducing per-unit onboarding from hours to under a minute.

Projects

PitWall — F1 Telemetry Dashboard | *Next.js, TypeScript, React Three Fiber*

- Built a real-time F1 telemetry dashboard with 3D track visualization (React Three Fiber) driven by a live Server-Sent Events stream, implementing arc-length parameterization for accurate car positioning and automatic SSE reconnect handling.
- Repaired and redeployed the app after diagnosing a car-positioning drift bug, hardening the SSE connection against dropped streams.

J.A.R.V.I.S. | *Python, FastAPI, Ollama, ChromaDB*

- Built a fully local voice AI: Porcupine wake-word, Whisper.cpp STT, Ollama LLM, Mem0 + ChromaDB RAG memory, and tool integrations for Gmail, Calendar, browser automation, and Telegram behind a FastAPI + WebSocket dashboard.
- Implemented persistent long-term memory via Mem0 + ChromaDB retrieval, enabling context-aware responses across sessions.

Smart Scheduler | *Python, FastAPI, React, TypeScript, Docker, OAuth 2.0*

- Architected a greedy deadline/priority scheduling backend with 120+ Pytest unit tests (85% coverage); containerized via Docker Compose with Google OAuth 2.0.
- Built a type-safe React + TypeScript frontend consuming a documented REST API, deployed via Docker Compose alongside the backend service.

RecruIT | *Node.js, TypeScript, React*

- Built a full-stack job-application tracking tool with a Node.js/TypeScript backend and a React/Vite frontend for managing application pipelines end-to-end.
- Modeled application stages (applied, interview, offer) with status tracking and reminders to reduce manual spreadsheet upkeep.

Activities

UT IEEE RAS · TSA Co-Founder (3× National Finals) · 4× Gold PVSA Medalist · SAMPADA Certified Violinist, ABRSM Grade 5